Supplementary Materials for ADMET-Bench: Task-Dependent Performance of GNNs and Pretrained Models in ADMET Prediction

I. Architecture-Selection Experiments

This section reports the preliminary architecture-selection screen used to choose the graph neural network backbone. Eight message-passing variants (GraphConv, GCN, GAT, GraphSAGE, GIN, TAG, SGC, and TransformerConv) were evaluated on all six benchmark endpoints under the same scaffold-split pipeline, target transformations, and evaluation metrics used in the main study. For each architecture and endpoint the reported result is the best of five configurations selected on the validation split (search over hidden dimension, number of layers, and learning rate), giving $8 \times 6 \times 5 = 240$ training runs in total.

TABLE S1
Preliminary classification architecture screen: validation-split AUC-ROC and F1 (the model-selection basis), each architecture at its best of five validation-selected configurations. Best validation AUC per task in bold. Test-set performance was not used for architecture selection.

Architecture	Tox21 (NR-AR)		hERG	
	Val AUC	Val F1	Val AUC	Val F1
GraphConv	0.819	0.480	0.918	0.903
GCN	0.821	0.435	0.912	0.915
GAT	0.819	0.500	0.907	0.871
GraphSAGE	0.854	0.421	0.903	0.918
GIN	0.844	0.500	0.884	0.933
TAG	0.821	0.444	0.892	0.897
SGC	0.829	0.500	0.920	0.875
Transformer	0.778	0.480	0.905	0.881

TABLE S2
Regression architecture screen: average rank across the four ADME regression datasets by validation R^2 (lower rank is better), the basis on which the backbone was selected; each architecture at its best of five validation-selected configurations. Per-dataset validation R^2 shown; the final column gives mean test R^2 as a post-hoc reference only (not used for selection). GraphConv (deployed backbone) in bold.

Arch.	Avg Rank	Caco-2	Half-Life	Clear. Hep.	Clear. Mic.	Test R^2	Params	Time (s)
GraphSAGE	3.00	0.597	0.434	0.316	0.175	0.096	662k	7.3
Transformer	3.25	0.606	0.335	0.327	0.208	−0.197	316k	16.4
TAG	4.25	0.582	0.407	0.228	0.279	0.226	1059k	12.2
GraphConv	4.25	0.575	0.460	0.267	0.081	−0.673	81k	7.1
GCN	4.75	0.534	0.447	0.247	0.137	−0.673	399k	12.8
GAT	5.00	0.513	0.483	0.233	0.122	0.128	1067k	14.9
SGC	5.50	0.560	0.426	0.194	0.159	0.006	463k	12.6
GIN	6.00	0.625	0.368	0.157	0.058	−1.572	113k	6.2

No architecture dominates. Ranked by validation performance (the only leak-free selection signal), the eight backbones span a narrow average regression rank (3.0–6.0) with small per-dataset differences, and every architecture is unstable, with several collapsing to large negative test R^2, on the weakly-learnable clearance and half-life endpoints. This validation ordering is a poor predictor of test performance: the best validation-ranked backbone (GraphSAGE) reaches only mean test $R^2 = 0.10$, while several validation-competitive backbones post negative mean test R^2 (down to -1.57). GraphConv is mid-ranked on validation (regression validation rank 4.25 of 8) and is not a test-maximizing choice (its mean test R^2 is among the lowest); it was adopted as a single representative backbone, and because architecture selection used validation metrics only, the main-benchmark test evaluation remains unbiased.

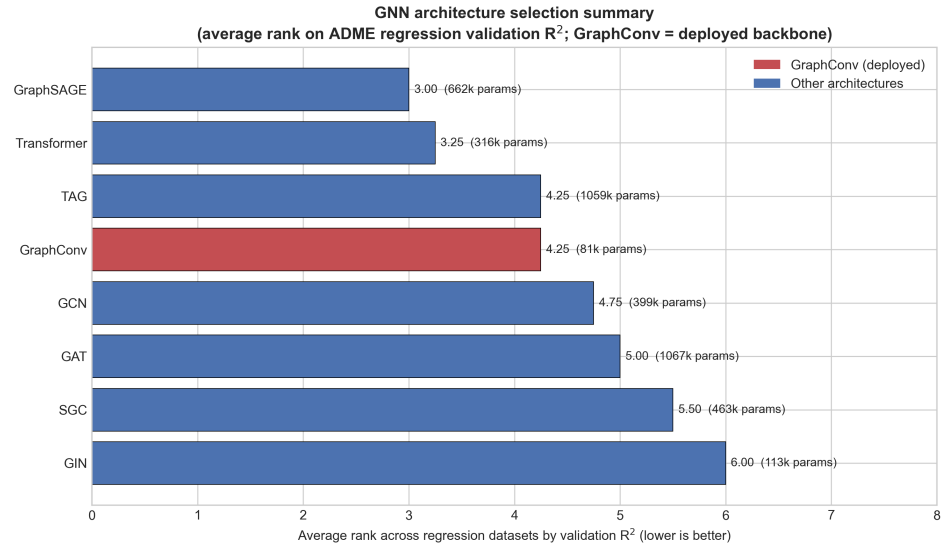


Fig. S1. Preliminary architecture-selection summary showing average rank across the regression datasets by validation R^2 (lower is better); GraphConv (the deployed backbone) is highlighted. No architecture dominates and the per-dataset differences fall within the noise of these small datasets; GraphConv was adopted as a single representative backbone using validation metrics only.

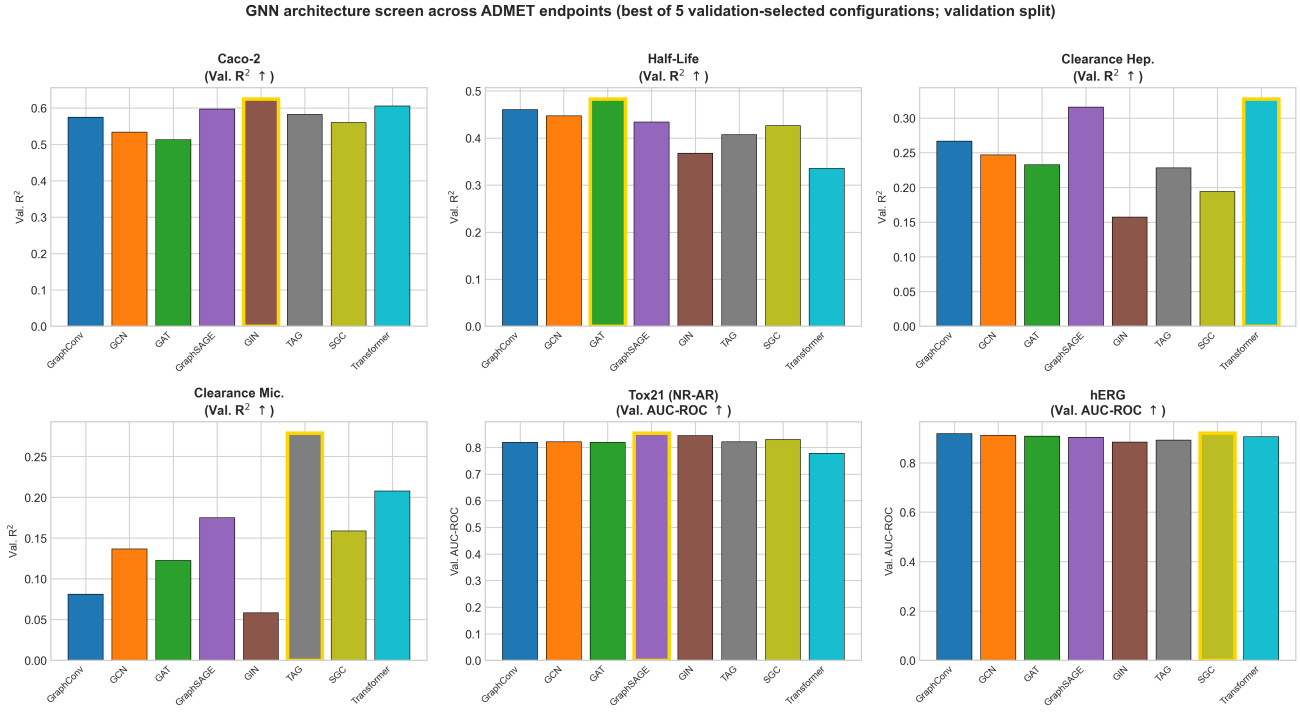


Fig. S2. Per-dataset architecture comparison across all six benchmark endpoints (classification by validation AUC-ROC, regression by validation R^2; the selection basis). The weakly-learnable clearance and half-life endpoints are unstable across all eight architectures, whereas the structure-driven endpoints cluster within a narrow band.

TABLE S3

HPO algorithms vs. Random Search (clean, leak-free re-run), each optimizer's validation-selected configuration evaluated once on test across the $n = 6$ datasets. Because the per-endpoint test metrics live on incomparable scales (R^2, occasionally pathologically negative, and AUC), we report direction-based and scale-controlled measures rather than a raw mean R^2/AUC difference. W/T/L: wins/ties/losses vs. Random Search. Sign-test p: exact two-sided binomial test on wins-vs-losses (ties dropped). Norm. effect: per-endpoint (optimizer−random) difference divided by the spread of all seven optimizers on that endpoint, averaged over the six endpoints (a scale-free, direction-normalized paired effect). Sensitivity columns recompute the mean paired difference with R^2 clipped to $[-1, 1]$ and floored at 0 (AUC untouched). All 95% CIs are 10,000-resample percentile bootstraps over the six paired differences. No optimizer is significant under the sign test, and every scale-free / clipped interval spans zero; only the aggressively compressed floor-0 variant shows two marginal effects (ABC slightly +, SA slightly −), neither indicating a consistent winner. So none consistently outperforms Random Search under the 50-trial budget. Full per-trial histories are released with the code.

Optimizer	W/T/L	Sign p	Norm. effect [95% CI]	Clip $[-1, 1]$ mean [95% CI]	Floor-0 mean [95% CI]
PSO	2/0/4	0.69	−0.16 [−1.35, +1.25]	+0.174 [−0.084, +0.612]	+0.033 [−0.036, +0.124]
ABC	5/0/1	0.22	+0.52 [−0.45, +1.38]	+0.026 [−0.292, +0.311]	+0.032 [+0.000, +0.091]
GA	3/0/3	1.00	+0.01 [−1.41, +1.40]	+0.169 [−0.067, +0.536]	+0.020 [−0.040, +0.086]
SA	1/0/5	0.22	−0.77 [−1.82, +0.55]	−0.135 [−0.471, +0.171]	−0.018 [−0.031, −0.006]
HC	3/0/3	1.00	+0.02 [−1.13, +1.37]	+0.034 [−0.440, +0.556]	+0.014 [−0.021, +0.062]
TPE	2/0/4	0.69	+0.01 [−1.02, +1.16]	+0.105 [−0.075, +0.329]	+0.027 [−0.024, +0.114]

TABLE S5

HPO-algorithm comparison (clean re-run): test performance of each optimizer's validation-selected configuration across datasets (regression R^2, classification AUC-ROC). Shown for transparency; optimizers are compared statistically against Random Search in Table S3, and no optimizer is ever selected using test data. Regression R^2 is highly unstable on the weakly-learnable clearance endpoints, so per-optimizer regression values should not be over-interpreted.

Dataset	PSO	ABC	GA	SA	HC	Random	TPE
Regression (R^2 ↑)							
Caco2_Wang	0.508	0.553	0.450	0.519	0.519	0.552	0.515
Half_Life_Obach	0.119	0.211	0.193	−0.285	0.037	0.035	0.274
Clearance_Hep.	−0.269	−0.762	−0.219	−1.351	−0.989	−0.114	−0.283
Clearance_Mic.	0.235	−0.387	0.058	−0.529	0.130	−3.288	−0.365
Classification (AUC-ROC ↑)							
Tox21 (NR-AR)	0.739	0.757	0.730	0.739	0.758	0.743	0.726
hERG	0.738	0.810	0.830	0.771	0.777	0.810	0.787

II. Hyperparameter-Optimization Details

TABLE S4

Hyperparameter search space for GNN models. The three MLP-head layer widths are sampled independently and then sorted in descending order (Layer 1 ≥ Layer 2 ≥ Layer 3) to yield a funnel-shaped prediction head.

Hyperparameter	Search Space
Hidden Dimensions	{64, 96, 128, 192, 256, 384, 512}
Number of Layers	{3, 4, 5, 6, 7}
Learning Rate	[1e-4, 1e-2] (log-uniform)
Weight Decay	[1e-6, 1e-2] (log-uniform)
MLP Head Layer 1	{128, 192, 256, 384, 512}
MLP Head Layer 2	{64, 96, 128, 192, 256}
MLP Head Layer 3	{32, 48, 64, 96, 128}

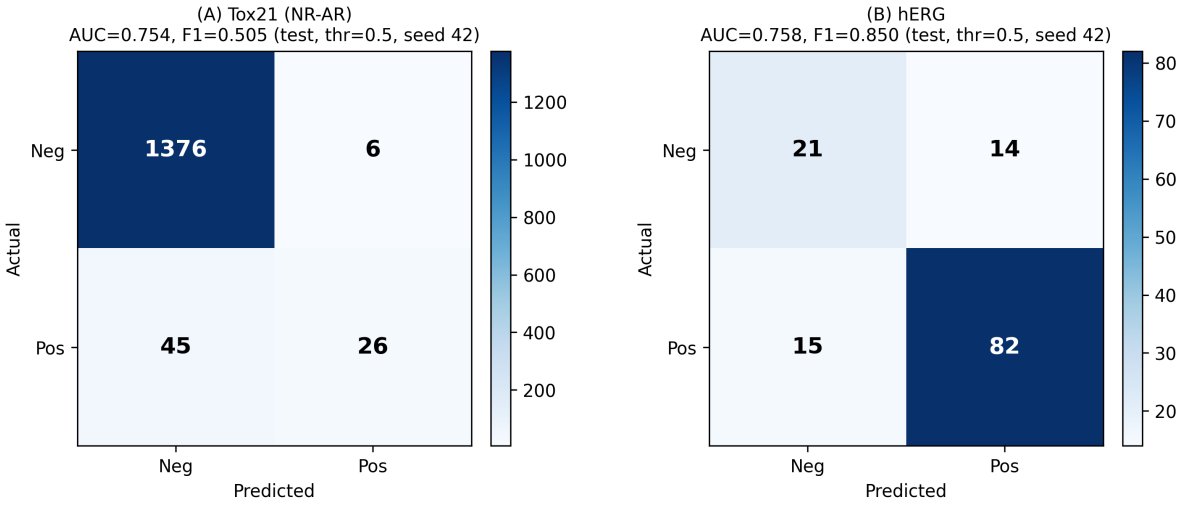


Fig. S3. Confusion matrices for the GNN's validation-selected configuration on the two toxicity tasks, on the held-out test set at the standard 0.5 decision threshold (seed 42). (A) Tox21 shows majority-class bias due to extreme imbalance (F1 = 0.51). (B) hERG shows balanced performance (F1 = 0.85). Across-seed, threshold-tuned aggregates (mean over ten seeds) are reported in Fig. S4 and the imbalance-metric text.

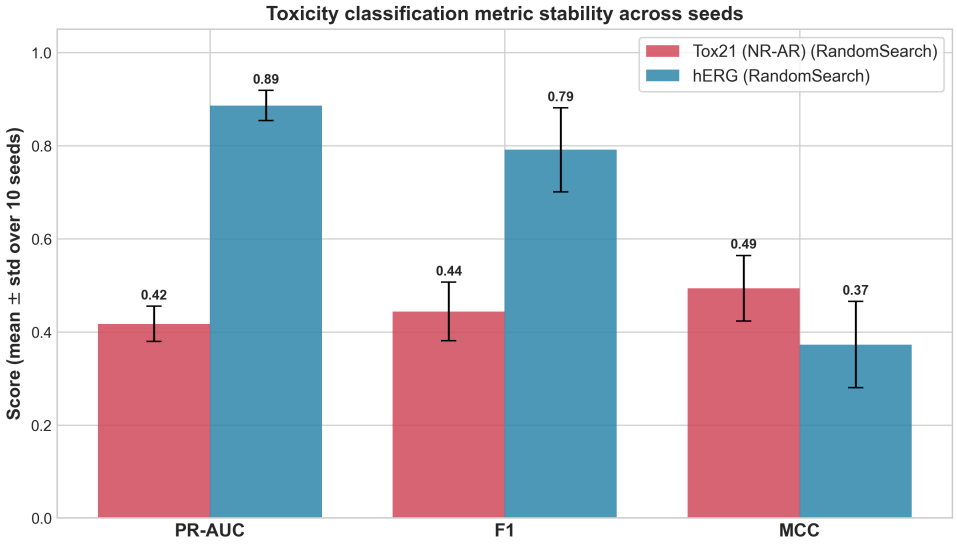


Fig. S4. Across-seed stability of the imbalance-aware toxicity metrics (PR-AUC, F1, MCC). Bars show the mean over ten seeds and error bars show ± one standard deviation. These imbalance-aware summaries use mean ± standard deviation to quantify across-seed variability; the primary AUC-ROC results are summarized using median [min–max] in the main manuscript. Across-seed variability is modest for both endpoints, with the threshold-dependent F1 and MCC somewhat more variable on hERG than on Tox21.